XAI1

Ontology review

Criterion	Statement	Rate from 1-5
XAI domain adequacy	The ontology adequately represents the XAI field	4
Completeness	The ontology covers the most important XAI concepts.	4
Explanation usefulness	The ontology supports understanding AI explanations.	3
Explanation relevance	The represented explanation types are meaningful and useful.	4
CQ support	The ontology outputs correspond to realistic information needs.	3
Stakeholder awareness	Stakeholder-specific explanation needs are represented appropriately.	3
Explanation diversity	The ontology adequately distinguishes different explanation approaches.	4
Explanation quality representation	Explanation quality criteria are represented appropriately.	3
Explainability coverage	The ontology covers the major explainability concerns relevant to this use case.	3.5
Explainability terminology	Explanation concepts and definitions are used consistently and correctly.	5
Actionability support	The ontology captures actionable explanation knowledge.	3
Future applicability	The ontology can accommodate future XAI methods and explanation types.	5

Overall Assessment

Criterion	Statement	Rate from 1-5
Overall adequacy	The ontology adequately represents explainable machine-learning pipelines.	**3.5**
Stakeholder support	The ontology adequately supports stakeholder-specific explanations.	**3.5**
Conceptual completeness	The ontology captures the most important concepts of the domain.	**4**
Practical usefulness	The ontology would be useful in practice.	**4**
Reusability	The ontology could be reused in other explainable AI contexts.	**4**
Overall quality	The ontology is of sufficient quality for its intended purpose.	**3.5**

Issue list

ID	Issue Description	Proposed Modification
I1	Support for emerging AI systems and their explanations is incomplete.	Extend the ontology with concepts relevant to deep learning, transformers, LLMs, and explanation methods for such models.
I2	The ontology provides limited support for alternative and human-centered explainability evaluation approaches.	Support additional explainability evaluation frameworks and perspectives, particularly human-centred evaluation approaches.
I3	Stakeholder explanation interactions are represented too statically.	Extend the ontology to accommodate interactive explanations and follow-up questioning.

XAI2

Ontology review

Criterion	Statement	Rate from 1-5
XAI domain adequacy	The ontology adequately represents the XAI field	4
Completeness	The ontology covers the most important XAI concepts.	5
Explanation usefulness	The ontology supports understanding AI explanations.	3
Explanation relevance	The represented explanation types are meaningful and useful.	-
CQ support	The ontology outputs correspond to realistic information needs.	4
Stakeholder awareness	Stakeholder-specific explanation needs are represented appropriately.	5
Explanation diversity	The ontology adequately distinguishes different explanation approaches.	5
Explanation quality representation	Explanation quality criteria are represented appropriately.	3
Explainability coverage	The ontology covers the major explainability concerns relevant to this use case.	4
Explainability terminology	Explanation concepts and definitions are used consistently and correctly.	5
Actionability support	The ontology captures actionable explanation knowledge.	3
Future applicability	The ontology can accommodate future XAI methods and explanation types.	3

Overall assessment

Criterion	Statement	Rate from 1-5
Overall adequacy	The ontology adequately represents explainable machine-learning pipelines.	4
Stakeholder support	The ontology adequately supports stakeholder-specific explanations.	3
Conceptual completeness	The ontology captures the most important concepts of the domain.	5
Practical usefulness	The ontology would be useful in practice.	4
Reusability	The ontology could be reused in other explainable AI contexts.	4
Overall quality	The ontology is of sufficient quality for its intended purpose.	5

Issue list

ID	Issue Description	Proposed Modification
I1	Screening nurses require explanations in the same format as existing screening workflows.	Explicitly represent explanation presentation format, interaction modality, or delivery channel within stakeholder-specific explanation requirements.
I2	The ontology is tied too closely to the Co-12 explanation quality framework.	Make explanation quality frameworks extensible and assess whether VXAI should be supported alongside Co-12.
I3	The ontology appears too complex and over- whelming for users.	Introduce a layered "telescope" presentation: high-level overview model, module-level views, and detailed OntoUML views.
I4	Stakeholder targeting of explanations is not represented explicitly enough.	Explicitly model explanation recipients and intended stakeholders, and include them in the RDF example.

I5	Covariate Complexity may not be meaningful for patients.	Reconsider assignment of Covariate Complexity as an explanation quality criterion for patients.

ONT1

Ontology review

Criterion	Statement	Rate from 1-5
Conceptual correctness	The concepts are represented correctly	**4**
Concept necessity	All introduced concepts are necessary	**5**
Stakeholder modelling	Stakeholder roles are represented appropriately	**5**
Type modelling	Explanation types are classified appropriately	**5**
Ontological completeness	All important ontology elements are represented	**5**
Foundational alignment	The OntoUML/gUFO grounding is appropriate	**-**
Semantic consistency	Relations and constraints are logically coherent	**5**
Constraint adequacy	Cardinalities, restrictions, and constraints are appropriate	**-**
Modular consistency	The extension is consistent with the original XMLPO ontology	**-**
Extensibility	The ontology can be extended without major redesign	**5**
Interoperability	The ontology sufficiently supports integration with external ontologies and knowledge graphs	**2**
Reuse quality	Reused ontologies are integrated appropriately	**2**
Documentation clarity	Definitions are understandable	**4**

Overall assessment

Criterion	Statement	Rate from 1-5
Overall adequacy	The ontology adequately represents explainable machine-learning pipelines.	5
Stakeholder support	The ontology adequately supports stakeholder-specific explanations.	5
Conceptual completeness	The ontology captures the most important concepts of the domain.	-
Practical usefulness	The ontology would be useful in practice.	5
Reusability	The ontology could be reused in other explainable AI contexts.	-
Overall quality	The ontology is of sufficient quality for its intended purpose.	4

Issue list

ID	Issue Description	Proposed Modification
I1	Support for AI agents is incomplete	Extend the stakeholder model with artificial agents.
I2	The label ExplanationArtifact is unclear.	Refine or rename the concept to better communicate its intended meaning.
I3	The ontology implementation is insufficiently developed for publication and reuse.	Provide an OWL implementation, example instances, namespace declarations, OWL imports, and repository publication.
I4	Semantic interoperability is insufficiently demonstrated.	Add an interoperability use case with data flow examples and evidence of semantic alignment across systems.
I5	The validation methodology could be more rigorous.	The validation methodology could be more rigorous.

I6	Ontology curation process is not defined.	The working version of the ontology can be uploaded to GitHub, which has an issue functionality that enables GitHub users to share their issues with it.

ONT2

Ontology review

Criterion	Statement	Rate from 1-5
Conceptual correctness	The concepts are represented correctly	**4**
Concept necessity	All introduced concepts are necessary	**5**
Stakeholder modelling	Stakeholder roles are represented appropriately	**4**
Type modelling	Explanation types are classified appropriately	**4**
Ontological completeness	All important ontology elements are represented	**5**
Foundational alignment	The OntoUML/gUFO grounding is appropriate	**5**
Semantic consistency	Relations and constraints are logically coherent	**5**
Constraint adequacy	Cardinalities, restrictions, and constraints are appropriate	**4**
Modular consistency	The extension is consistent with the original XMLPO ontology	**-**
Extensibility	The ontology can be extended without major redesign	**4**
Interoperability	The ontology sufficiently supports integration with external ontologies and knowledge graphs	**5**
Reuse quality	Reused ontologies are integrated appropriately	**3**
Documentation clarity	Definitions are understandable	**4**

Overall assessment

Criterion	Statement	Rate from 1-5
Overall adequacy	The ontology adequately represents explainable machine-learning pipelines.	4
Stakeholder support	The ontology adequately supports stakeholder-specific explanations.	4
Conceptual completeness	The ontology captures the most important concepts of the domain.	5
Practical usefulness	The ontology would be useful in practice.	5
Reusability	The ontology could be reused in other explainable AI contexts.	5
Overall quality	The ontology is of sufficient quality for its intended purpose.	4

Issue list

ID	Issue Description	Proposed Modification
I1	The ontology attempts to represent process-related phenomena using OntoUML constructs, although some aspects may be better captured using a dedicated process modelling language.	Reassess process-oriented portions of the ontology and either simplify them or complement them with an appropriate process modelling formalism.
I2	The ontology diagrams are overly large and difficult to read, limiting comprehensibility and reviewability.	Replace full-model views with multiple smaller module-specific diagrams and reorganize the model into clearly separated packages.
I3	Relationship names such as *Supports* and *Handles* are overly generic and do not adequately communicate domain semantics.	Rename associations using more precise domain-specific relationship names.

I4	Several cardinalities appear overly permissive (e.g., 0..*), potentially concealing important domain constraints.	Revisit multiplicities and introduce stricter cardinality constraints where justified by the domain.
I5	Cyclic dependencies are present without explicit formal constraints to ensure semantic consistency.	Introduce OCL or first-order logic constraints to justify and constrain cyclic structures.
I6	The stakeholder module contains numerous role classes whose distinctions are not clearly justified from an ontological perspective.	Define identity and classification rules for stakeholder roles and retain only roles supported by clear ontological differences.
I7	The modelling choice of representing explanation goals as subkinds may be inappropriate because the categories appear to function as values rather than object types.	Evaluate replacing explanation-goal subkinds with an enumeration or individual instances of a goal category.
I8	The relation between ExplanationGoal and XAIStakeholder may not be adequately grounded through explicit material and relator-based relations.	Remodel the dependency using a material relation supported by an appropriate relator, following UFO/OntoUML guidelines.
I9	The explanation type module contains a multi-level modelling problem by mixing classification levels.	Separate explanation-type individuals from explanation-artifact classes and explicitly model the classification relation between them.
I10	The ontology does not clearly distinguish whether quality dimensions and perspectives are classes, categories, or instances.	Establish a consistent ontological commitment and explicitly define the ontological status of dimensions and perspectives.
I11	The semantics of the *dimensionOf* relation are unclear.	Define the intended meaning of the relation and consider modelling it as a composition (*componentOf*) relationship if appropriate.
I12	The relationship between explanation-quality dimensions and	Explicitly model stakeholder-dependent perspectives using relators, modes,

	stakeholder perspectives is insufficiently specified.	property chains, or other suitable OntoUML constructs.
I13	A separate hierarchical taxonomy view of explanation-quality dimensions is missing, making the structure difficult to understand.	Create an additional diagram focused solely on the taxonomy from root concepts to leaf concepts.
I14	Participation relations in the operational ontology are incorrectly represented using instantiation semantics.	Replace the instantiation relation with appropriate specialization/generalization relationships between participation relations.
I15	Some creation-related relations appear to require explicit redefinition semantics.	Introduce OntoUML «redefines» stereotypes where lower-level relations specialize more general relations.
I16	Certain manifestation relations may conceal intermediate events that are not explicitly represented.	Evaluate whether hidden event types should be explicitly modelled to clarify the semantics of manifestation.
I17	Directionality of associations is not always explicit, reducing semantic clarity.	Show navigability and relation direction consistently throughout the ontology diagrams.
I18	The relationship between MetabolicDisorder and DiabeticNeuropathy may be missing or under-specified.	Investigate the domain dependency and introduce an explicit relation if ontologically justified.
I19	The integration strategy for reused ontologies and domain ontologies remains unclear.	Explicitly document ontology reuse decisions and validate compatibility through OWL imports and reasoning.
I20	Naming conventions currently resemble data-model notation (camel case) rather than conceptual modelling notation.	Adopt space-separated conceptual labels to improve readability and conceptual clarity.
I21	Potential OntoUML anti-patterns have not been systematically assessed.	Perform anti-pattern detection and correct identified OntoUML modelling issues.
I22	The ontology has not been validated through population testing.	Use Alloy or similar tools to generate example populations and verify model consistency.

I23	The ontology may not adequately support representation needs of end users such as patients and screening nurses because conceptual models and user-facing views are conflated.	Distinguish the conceptual ontology from application-level presentation models and interface requirements.

ONT3

Ontology review

Criterion	Statement	Rate from 1-5
Conceptual correctness	The concepts are represented correctly	**3**
Concept necessity	All introduced concepts are necessary	**3**
Stakeholder modelling	Stakeholder roles are represented appropriately	**5**
Type modelling	Explanation types are classified appropriately	**3**
Ontological completeness	All important ontology elements are represented	**4**
Foundational alignment	The OntoUML/gUFO grounding is appropriate	**4**
Semantic consistency	Relations and constraints are logically coherent	**3**
Constraint adequacy	Cardinalities, restrictions, and constraints are appropriate	**4**
Modular consistency	The extension is consistent with the original XMLPO ontology	**-**
Extensibility	The ontology can be extended without major redesign	**4**

Interoperability	The ontology sufficiently supports integration with external ontologies and knowledge graphs	3
Reuse quality	Reused ontologies are integrated appropriately	3
Documentation clarity	Definitions are understandable	-

Overall assessment

Criterion	Statement	Rate from 1-5
Overall adequacy	The ontology adequately represents explainable machine-learning pipelines.	3
Stakeholder support	The ontology adequately supports stakeholder-specific explanations.	4
Conceptual completeness	The ontology captures the most important concepts of the domain.	3
Practical usefulness	The ontology would be useful in practice.	4
Reusability	The ontology could be reused in other explainable AI contexts.	3
Overall quality	The ontology is of sufficient quality for its intended purpose.	4

Issue list

ID	Issue Description	Proposed Modification
I1	Evaluation modelling is incomplete.	Represent evaluation scores explicitly and model stakeholder participation in evaluation events.
I2	Several relations require revision.	Remove the relations from ExplanationGoal to XAIStakeholder and

		from ExplanationQuality to ExplanationEvaluation.
I3	The Priority classes may be incorrectly modelled as relators.	Consider remodelling them as modes and renaming them to end on PrioritySetting.
I4	Explanation-type modelling contains multi-level modelling inconsistencies.	Separate explanation artifacts subkinds from explanation type instances and introduce the classifiedAs relation between them.
I6	Explanation dimensions and perspectives may be assigned inappropriate stereotypes.	Consider whether dimensions and perspectives should be modelled as modes or mental moments rather than kinds.
I7	Interoperability and reusability claims are difficult to assess from the current artefacts.	Provide additional implementation artefacts, examples, or evaluation evidence to support interoperability and reuse claims.

HC1

Ontology review

Criterion	Statement	Rate from 1-5
Domain adequacy	Reflects CAN screening practice	**5**
Clinical relevance	The represented concepts are meaningful in practice	**5**
Clinical correctness	Concepts are represented in a clinically correct manner	**4**
Terminology clarity	The terminology is understandable to healthcare professionals	**4**
Completeness	Important clinical concepts are modelled	**4**

Information sufficiency	Is enough information represented for the intended use case	5
Stakeholder distinction	Distinguishes stakeholders appropriately	5
Workflow alignment	The ontology reflects the screening workflow adequately	5
Usefulness	Supports understanding of screening outcomes and explanations?	5
Decision-support value	The represented information supports clinical decision-making	-
Practical applicability	The ontology can be realistically adopted in practice	-

Overall assessment

Criterion	Statement	Rate from 1-5
Overall adequacy	The ontology adequately represents explainable machine-learning pipelines.	-
Stakeholder support	The ontology adequately supports stakeholder-specific explanations.	5
Conceptual completeness	The ontology captures the most important concepts of the domain.	5
Practical usefulness	The ontology would be useful in practice.	5
Reusability	The ontology could be reused in other explainable AI contexts.	-
Overall quality	The ontology is of sufficient quality for its intended purpose.	4

Issue list

ID	Issue Description	Proposed Modification

I1	The role of the physician and the collaborative workflow with nurses are missing.	Explicitly model the physician's diagnostic role and the nurse-physician workflow.
I2	Standardized values lack context, and feature names do not match the target audience's knowledge.	Provide context for standardized values and adapt feature names to the target audience.
I3	Explanations lack clinical interpretations like population comparisons, missing data analysis, and outlier identification.	Incorporate population comparisons, missing data handling, and clinical outlier identification.